

Puspa Kamal Rai

9th Mile, Thurbo, Mirik, Darjeeling - 734214, West Bengal | rpuspakamal@gmail.com | +91 90023 34495

Website : puspa-opal | LinkedIn : Puspa Kamal Rai | GitHub : puspkml

Vision

Driven by dedication and consistent effort, I believe challenges can be overcome with the right mindset. I enjoy collaborating with people from diverse backgrounds and value the learning that comes from such experiences. My work is guided by Bhagawan Sri Sathya Sai Baba, who motivates me to continually push my limits and grow keeping my morals intact.

Career Objective

I have a deep interest in semiconductor manufacturing and electronics, but I am equally open to all domains and opportunities that help me to shape my skills. With knowledge in circuit design, embedded systems, FPGA, semiconductor manufacturing knowledge, coding, managing datasets, and automation, I enjoy solving problems and take pride in working independently and along with a team. I am adaptive, passionate, and curious to thrive in dynamic environments where I can put my domain knowledge and skills to use. Committed to continuous learning and growth, I am looking for real life problem statements, where I can make meaningful contributions while further developing my abilities.

Education

Sri Sathya Sai Institute of Higher Learning; M.Sc. in Physics 2025 - 2027(expected)

- Semester 2 ongoing (CGPA: 8.0)
- **Coursework:** Classic Mechanics, Classical Electrodynamics, Quantum Mechanics – I & II, Mathematical Physics, Experimental Methods in Physics – I & II, Electronics and Microcontroller Lab, Nuclear and Particle Physics, Modern Optics, Statistical Mechanics, Software Lab (SciLab and Molecular Dynamics)

Sri Sathya Sai Institute of Higher Learning; B.Sc. (Hons.) in Physics 2022 – 2025

- CGPA: 8.0
- **Coursework:** Classical Mechanics, Electromagnetism, Optics, Thermal Physics, Solid State Physics, Nuclear Physics, Quantum Mechanics, Atomic and Molecular Spectroscopy, Electronics-I and II, Microprocessors, Python Programming-I, Computational Techniques in Physics, Laboratory courses in Electronics, Mechanics, Optics, Electromagnetism, General Physics, Microprocessors, and Python Programming-II.

Snowdrops School; CBSE 10th and 12th 2020 and 2022

- 10th: 90.4%, 12th: 88.0%

Internships and Projects

Automatic Bell System, Self-Taken Project Mar 2025 – July 2025

- With motivation to return gratitude to the institute where I am currently studying, I undertook this project.
- Developed an automated wireless alarming bell system to ensure seamless adherence to schedules and emergency needs.
- Integrated ESP01 Wi-Fi module and relay module, and built an interactive web interface using Svelte Kit for schedule management via server-side actions.

Water Purity Detection and Microbial Fuel Cells, Open-Ended Project Feb 2025 – June 2025

- Improved water purity detection efficiency using DREAM Assay by building a fully automated low-cost spectrophotometer with Arduino.
- Microbial fuel cells, and secondary cultures were prepared and the activity was monitored using this spectrometer.

IIT Kharagpur, Internship May 2025 – July 2025

- Under the guidance of Dr. Shiva Kiran Bhakta, developed a compact, low-cost, and efficient spectrometer using 3D-printed components and a smartphone camera as a sensor.

- Designed the system to incorporate photonic crystals and Bloch surface waves (BSW) for advanced spectral analysis.
- Implemented a modular adapter system to allow flexible analysis of reference and sample spectra, providing an accessible platform for optical diagnostics and educational tools.

Heat Management Studies in Semiconductor Devices using COMSOL

Feb 2026 - April 2026

Multiphysics, M.Sc. Mini Project

- Studied how heat builds up and spreads in MOSFETs and PCB-based embedded systems using COMSOL Multiphysics 5.5, and identified key issues like hotspot formation and poor heat dissipation.
- Simulated air cooling and thermoelectric cooling methods and compared their effectiveness to understand the best way to keep devices running safely and efficiently.

Technologies

Languages: Python, Svelte Kit, HTML, CSS, JavaScript, SciLab, LaTeX, Matlab

Technologies: Arduino, Raspberry Pi, ESP8266, ESP32, LabView, LT-Spice, SQL, Power BI, MS based tools, COMSOL Multiphysics, 3D Printing, Physics analysis devices.

Achievements and Extracurriculars

Flutist and Piccolo Player, University Brass Band (Rhythmic Energy)

2022 – Forever

- Performed as a member of the University Brass Band, contributing to live performances at a variety of events under tight timelines. Collaborated closely with team members to deliver high-quality performances, strengthening skills in teamwork, time management, and meeting deadlines.

Vice Captain, Shanti House, Sri Sathya Sai Institute of Higher Learning

2024-2025

- Among hundreds of students, being one of the few students to get a chance to lead hundred of students was an opportunity to learn as a leader. I used this opportunity to lead my house (Shanti House) to victory in the annual championship across all competitions.

Amaravati Quantum Hackathon, 2nd place, JNTUA Anantapur

September 23, 2025

- Being adaptive to the problem statement kept in front, I achieved 2nd place in the semi-finals by implementing the BB84 quantum key distribution protocol using LabVIEW.

Hostel Activities, Sri Sathya Sai Hostel for Senior Boys

September 23, 2025

- Hostel being a platform for community living, taking part in all the functions actively and being able to serve the place by any means as a gesture of gratitude and for my lord *Bhagawan Sri Sathya Sai Baba*.